ESERCIZIO

$V_S = 10V$

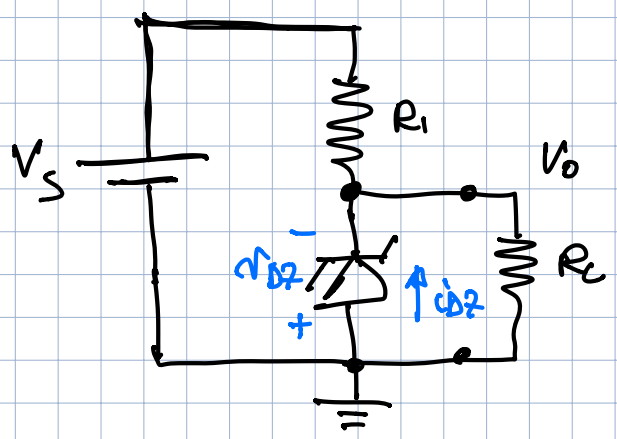
$R_1 = 2K\Omega$

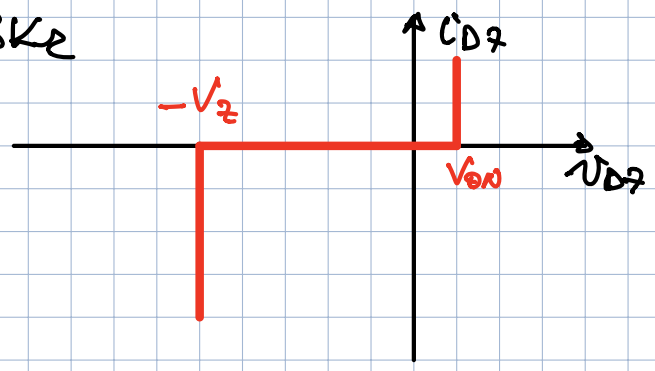
$R_L = 3K\Omega$

D7: $V_{ON} = 1V$

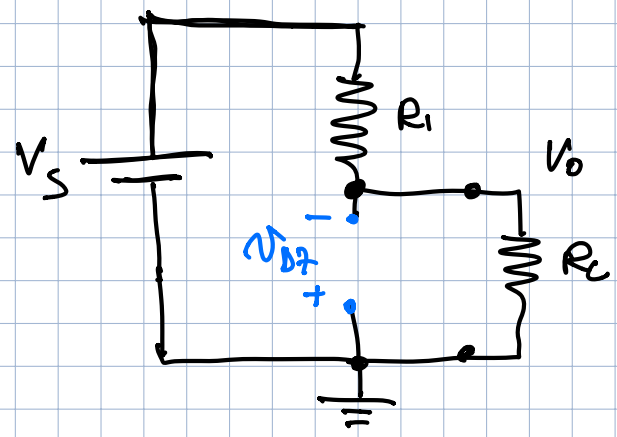
$V_Z = 5V$

$R_{ON} = R_Z = 0$





HP. D7 = OFF

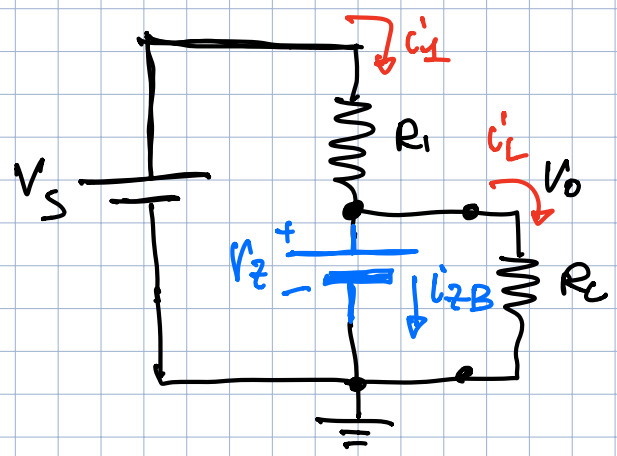


$$V_0 = V_S \cdot \frac{R_L}{R_1 + R_L} = 10V \cdot \frac{3K\Omega}{5K\Omega} = 6V$$

$$= v_{D7} = 0 - V_0 = -6V < V_Z$$

$\Rightarrow$ D7 = "BREAK"

SUPPONGO D7 IN BREAKDOWN



$$i_1 = \frac{V_S - V_Z}{R_1} = \frac{10V - 5V}{2K\Omega} = 2{,}5\,mA$$

$$i_L = \frac{V_Z}{R_L} = \frac{5V}{3K\Omega} = 1{,}67\,mA$$

$$i_{ZB} = i_1 - i_L = 0{,}8\bar{3}\,mA > 0$$

$\Rightarrow$ OK D7 = "BREAK"

$$V_0 = V_Z = 5V$$

MA SE TOLGO IL DIODO D7 e METTO $R_1 = 3K\Omega$

ALLORA $V_0 = V_S \cdot \frac{3K\Omega}{(3+3)K\Omega} = \frac{V_S}{2} = 5V$!

D7 è INUTILE?

CONSIDERO V_S VARIABILE DA 8V a 12V (V_S RUMOROSO)

$$i_{ZB} = i_1 - i_L = \frac{V_S - V_Z}{R_1} - \frac{V_Z}{R_L} > 0$$

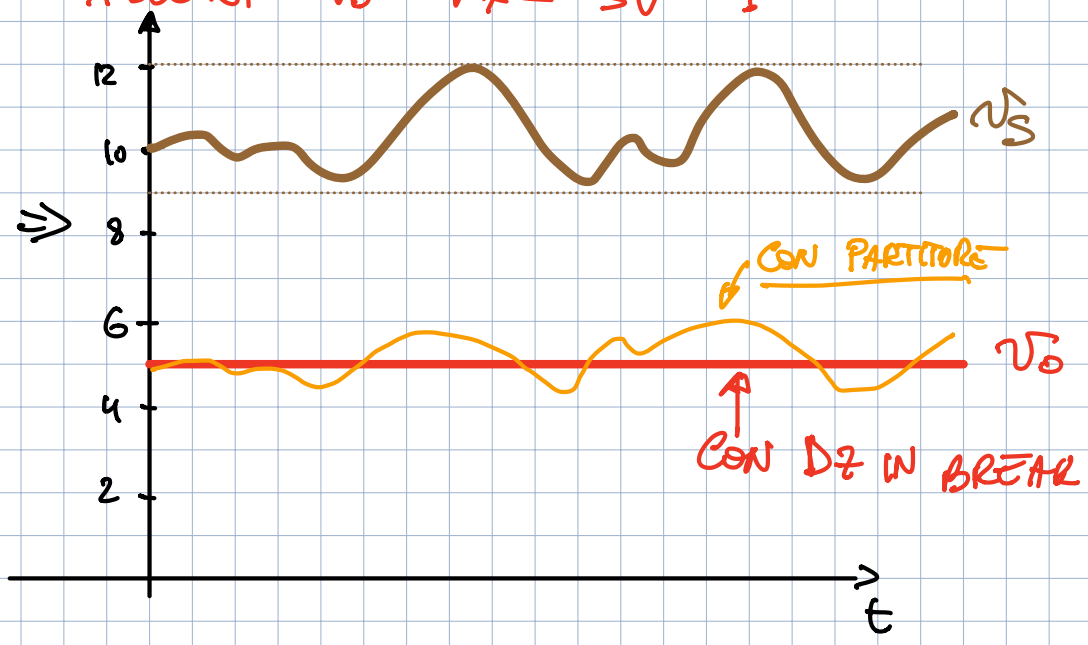
$$\frac{V_S}{R_1} > V_Z\left(\frac{1}{R_1} + \frac{1}{R_L}\right)$$

$$\Rightarrow V_S > V_Z \cdot \frac{R_1 + R_L}{\cancel{R_1} R_L} \cdot \cancel{R_1}$$

$$V_S > 5V \cdot \frac{5k\Omega}{3k\Omega} = 8.33V$$

CON $V_S > 8.33V$ ALLORA $i_{ZB} > 0$ e LO ZENER RESTA IN BREAK.

$\Rightarrow$ SE DZ RESTA IN BREAKDOWN ALLORA $V_O = V_Z = 5V$!



LO ZENER "STABILIZZA" LA TENSIONE

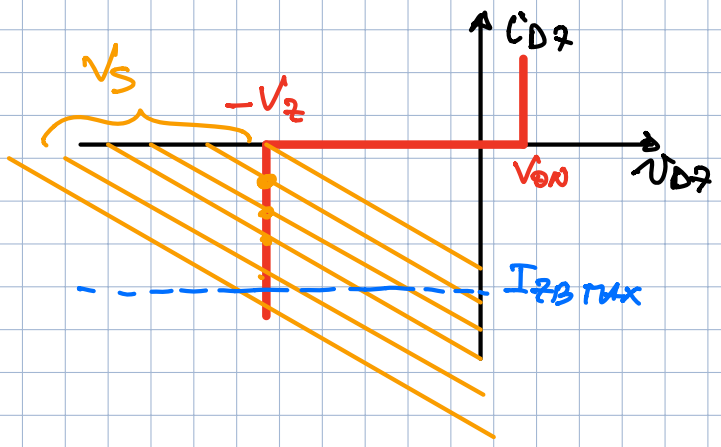
UNICA AVVERTENZA:

1) MANTENERE LO ZENER IN BREAKDOWN:

$\Rightarrow$ SE $V_S < 8.33V$

$\Rightarrow$ DZ = "OFF" e

$$v_O = V_S \cdot \frac{R_L}{R_1 + R_L} \;!$$



2) EVITARE CHE $i_{ZB} > i_{ZBMAX}$

CALCOLIAMO LA TRANSCARATTERISTICA:

HP $D_Z = OFF$ $\qquad v_{D_Z} = -V_S \dfrac{R_L}{R_1 + R_L}$

1) $v_{D_Z} > V_{ON} \Rightarrow D_Z = \text{"ON"} \Rightarrow -V_S \dfrac{R_L}{R_1+R_L} > V_{ON}$

$V_S < -V_{ON} \dfrac{R_1+R_L}{R_L} = 1V \cdot \dfrac{5}{3}$

$\Rightarrow V_S < -\dfrac{5}{3}V \Rightarrow D_Z = \text{"ON"}$

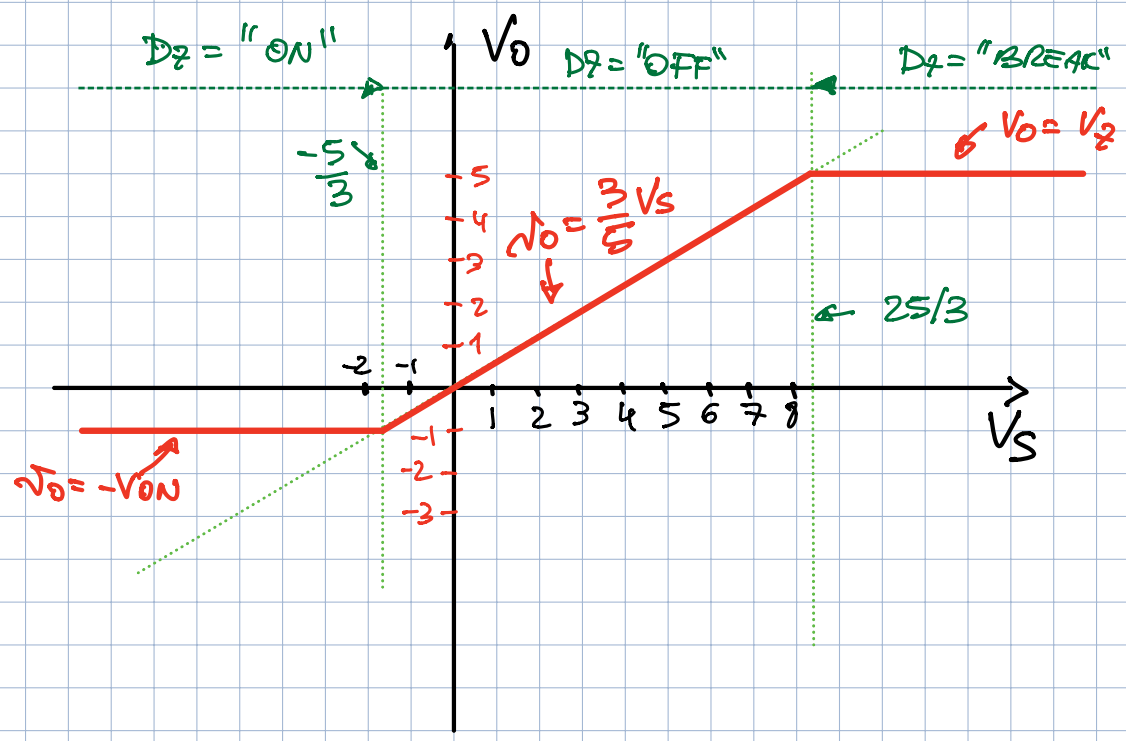
$\boxed{v_O = -V_{ON}\ !}$

2) $v_{D_Z} < -V_Z \Rightarrow D_Z = \text{"BREAK"} \Rightarrow -V_S \dfrac{R_L}{R_1+R_2} < -V_Z$

$V_S > V_Z \cdot \dfrac{R_1+R_2}{R_L} = 8{,}3\overline{3}\ V$

$\Rightarrow V_S > 8{,}3\overline{3} \Rightarrow D_Z = \text{"BREAK"}$

$\boxed{v_O = V_Z}$

3) $-\dfrac{5}{3}V \le V_S \le \dfrac{25}{3}$ $\qquad$ DIODO = OFF

$$\boxed{v_O = V_S \cdot \frac{R_L}{R_1+R_L} = V_S \cdot \frac{3}{5}}$$